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Mechanical Radial Drilling for Enhanced Oil Recovery in Mature Wells

Sajjad Akram, Rakesh Kumar, Dr. Faisal Derweesh, Razan Al Jabery, Faisal Salem Al Hosani, Emeka Val, Basma Banihammad, Ahmad Othman Alshaigy, and Mohamed H. Thani Al Hammadi, ADNOC Onshore, Abu Dhabi, UAE; Lee Dolman and Salavat Miftakhov, PGI Technologies Ltd., Abu Dhabi, UAE

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Abstract

Mechanical Radial Drilling (MRD) technology is designed to by-pass skin and wellbore damage zones and optimize reservoir contact. By augmenting the exposed reservoir surface area with multiple channels drilled with a 2 3/4" bit, production is enhanced significantly. For the first time ever in the target field, a candidate well with low permeability and low porosity was selected for this technology application. Technology allows for precise acidizing directly into the channels.

Mechanical Radial Drilling technology was used to create two channels in a candidate well in the target field. The anchor was set at 11367 ft, and the first channel, 33 ft in length, was drilled downward. The second channel, 20 ft in length, was drilled shallower in the same azimuth. Small-sized mud motors with a dogleg severity capability of 215 deg/100ft were used for accurate drilling. This approach bypassed damage zones, targeted specific intervals by crossing impermeable barriers, and allowed re-entering each channel for acid stimulation and trajectory surveys.

After drilling the first channel and confirming the trajectory with a survey, an acid treatment was performed using a special jet sub. 15 barrels of 15% HCL were pumped, resulting in good post-job injectivity at 2100 psi. The second channel also underwent acid treatment using a jet sub, with 30 barrels of 15% HCL pumped, showing promising post-job injectivity values at 2100 psi.

After acidizing the second channel, the well kicked at 9 ppg mud weight, initially inducing 3-4 bbl. losses. The well was stabilized by displacing/conditioning with 9.3 ppg mud. Injectivity values in both channels increased at the same injection pressure of 2100 psi. This indicated that the drilling technique successfully overcame near-wellbore damage zones, thus enhancing oil recovery from the candidate well.

These results indicate that the acid treatments were effective and improved acid injectivity, which is a positive outcome for enhancing oil recovery in the target field.

Mechanical Radial Drilling technology is unique and novel, using small-sized mud motors and conventional bits. It allows radial penetration of the reservoir with high accuracy, providing significant orthogonal displacement relative to the main wellbore at any desired orientation. The technology bypasses near-wellbore damage, connects previously unconnected layers, enters production zones with multiple large bore production channels that connect directly to the mother bore/production tubing for unrestricted flow

Introduction

The pursuit of enhanced oil recovery (EOR) techniques has been a pivotal focus within the petroleum industry, especially as the world's demand for energy continues to rise. One of the advanced methods of gaining traction in recent years is mechanical radial drilling, a procedure that holds promises for revitalizing mature wells and maximizing their output.

As part of the Operating company strategy, new tight reservoirs (<1 mD permeability) are being developed to fuel future production. However, conventional well stimulation techniques have fallen short, leading to rapid pressure decline and jeopardizing mandated targets. To bridge this gap, Operating Company evaluated the application of mechanical radial drilling in open-hole horizontal wells within a carbonate reservoir. The objective was to assess the feasibility and performance of drilling large-diameter radial production channels (2-3/4") to improve reservoir contact and enable precise acid placement. This approach aimed to enhance hydrocarbon recovery by combining mechanical channel creation with targeted acid jetting stimulation. The trial provided valuable insights into operational execution, stimulation effectiveness, and production response, contributing to the broader understanding of advanced well completion strategies in complex carbonate environments.

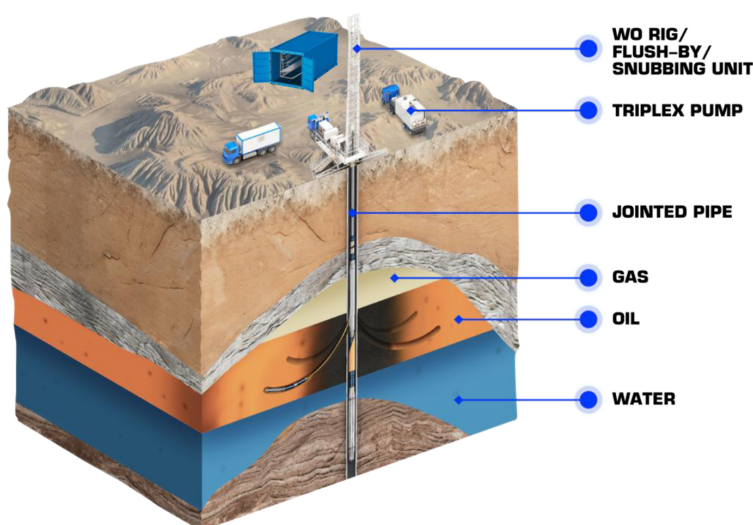


Figure 1—Mechanical Radial Drilling concept (04 channels drilled from vertical section)



Figure 2—Application of MRD in horizontal well

Background

The concept of EOR is not new; it encompasses a variety of techniques designed to extract the remaining oil from a reservoir after conventional methods have been exhausted. Traditional EOR methods include

thermal recovery, gas injection, and chemical flooding. However, these methods often come with high costs and complex operational requirements.

Mechanical radial drilling offers an innovative alternative. This technique involves creating multiple channels from the main wellbore, thereby enhancing the surface area in contact with the reservoir and facilitating unrestricted and improved oil flow.

Well Objectives

The primary objective of well ABC was to evaluate the technical feasibility and operational reliability of the Stimulation technology in drilling short-radius directional channels along a pre-defined trajectory within the geological framework of the XYZ field. Specifically, the plan aimed to drill four radial channels, each approximately 30 feet in length, at distinct measured depth intervals to access untapped reservoir zones located beneath the motherbore. Following channel creation, acid stimulation was to be performed to enhance reservoir connectivity and productivity. Although standard channels lengths typically extend to 45 feet, the use of the Kelly rig imposed mechanical constraints that limited channel length from the standard 45 ft possible with a top-drive or power swivel. These restrictions were primarily due to the limited Kelly length available during channels drilling, which affected both reach and downhole manipulation capabilities.

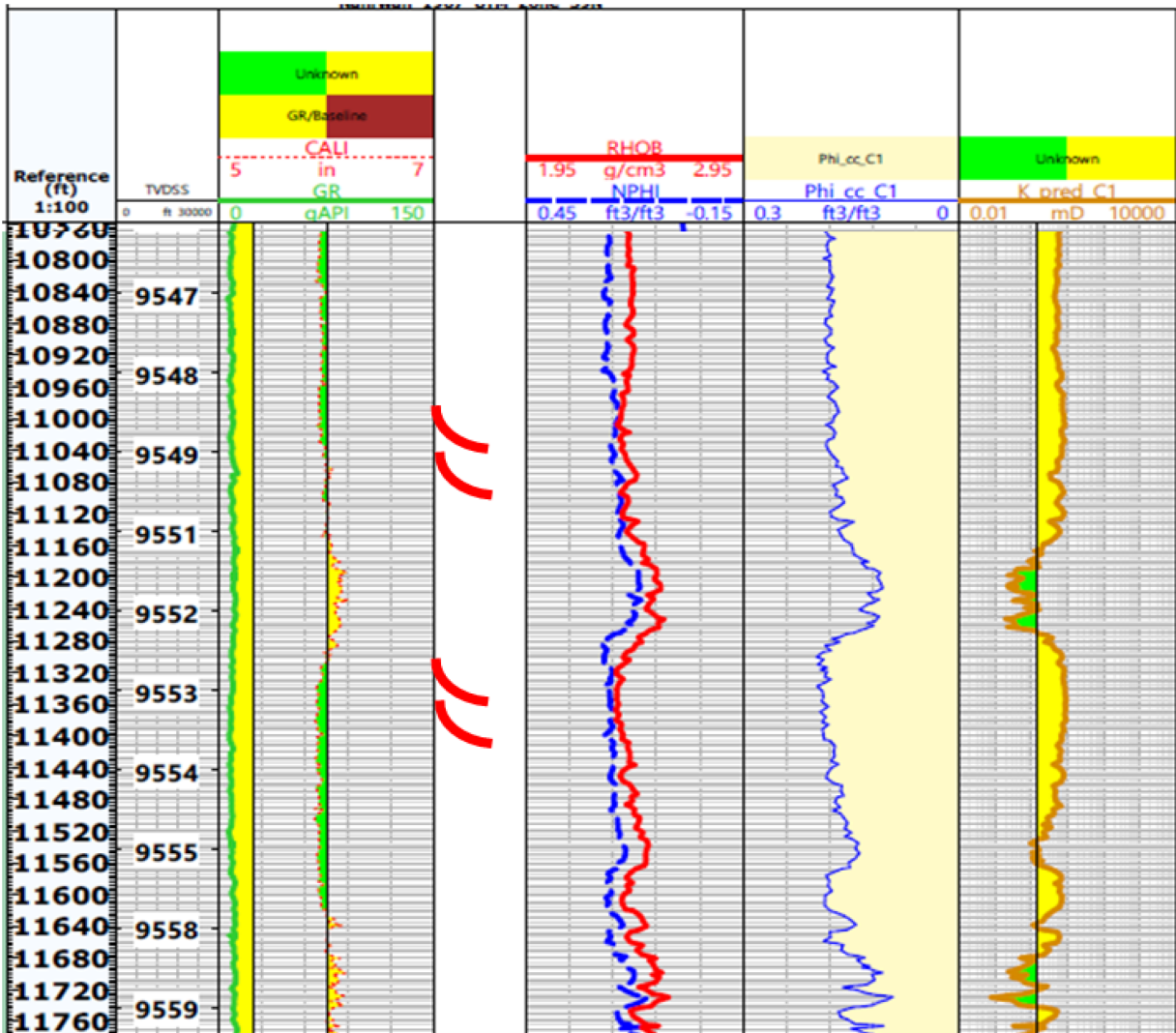


Figure 3—Open hole Logs (Channels orientation, length & position are selected based on logs)

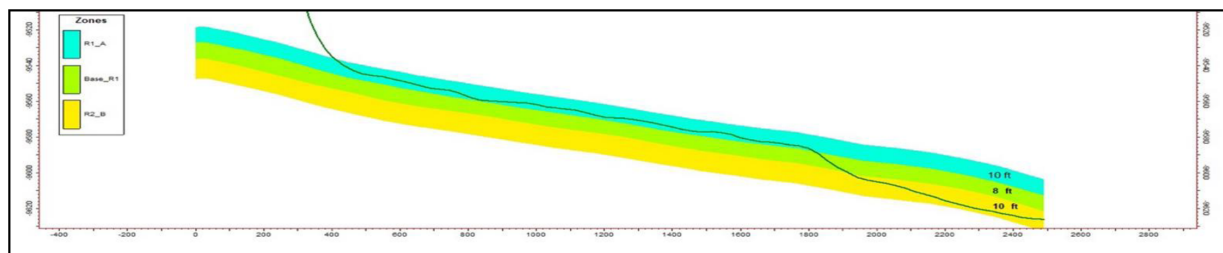


Figure 4—Chosen Well 2D plan (Actual well trajectory into different sub-layers)

Methodology and Technology

Mechanical Radial Drilling (MRD), represents a novel yet field-proven reservoir stimulation technique, offering a unique approach to lateral reservoir penetration. Though relatively new in widespread application, the technology is well-documented in literature (Lyagov et al., 2013–2022) and has demonstrated successful deployment in conventional oil and gas wells since 2019 (Galas et al., 2022; Bashirov et al., 2020; Basyrov et al., 2021).

What sets MRD technology apart is its ability to drill multiple large-diameter radial channels each 2-3/4 inches in diameter and 45 to 80 feet in length with high trajectory accuracy and minimal vertical deviation (± 3 ft). These channels extend horizontally from the main wellbore using a compact bottom-hole assembly (BHA) that integrates a whipstock, an anchor module with orienting capabilities, and a multi-section positive displacement motor (PDM) configured to deliver ultra-small-radius curved paths (typically >100 – $200^\circ/100$ ft).

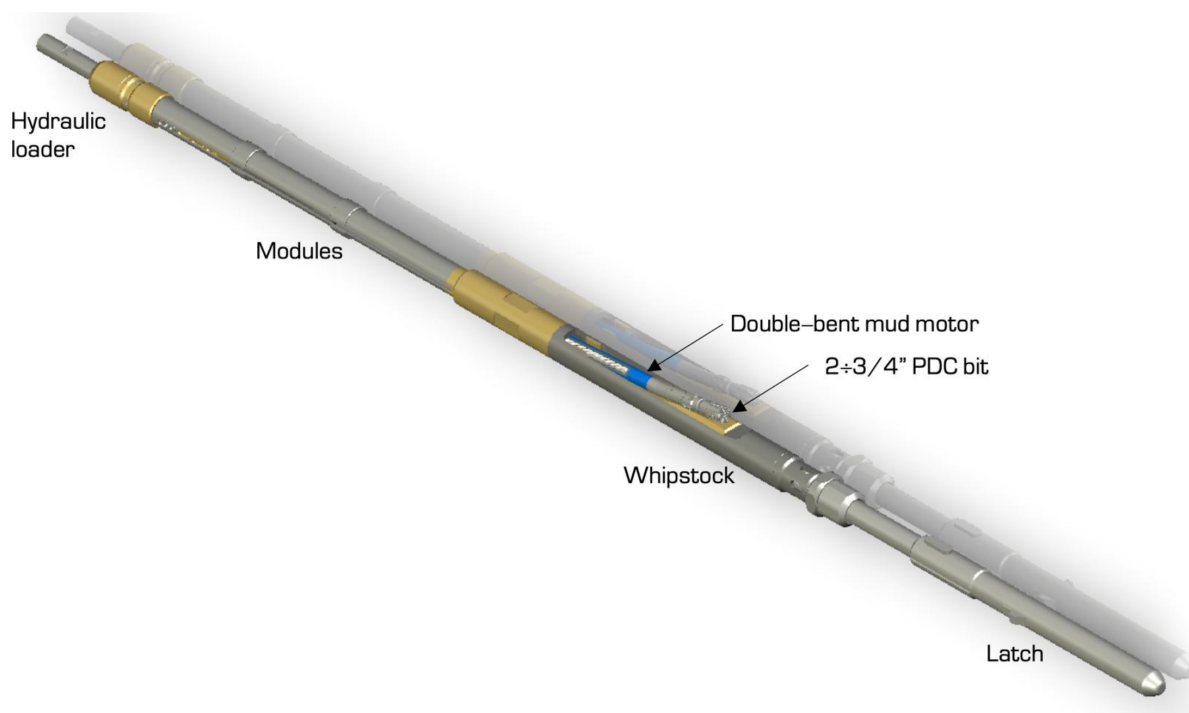


Figure 5—BHA for Mechanical Drilling

Unlike conventional stimulation methods, the technical system mechanically mills a window in the casing and directly penetrates the reservoir, bypassing near-wellbore damage and eliminating dependence on natural fracture systems or acid worm holing. The result is the creation of multiple direct-flow channels with unrestricted connectivity to the motherbore or production tubing. These laterals distribute drawdown

pressure across a wider radius, reducing the risk of early water or gas breakthrough typically caused by "coning" through traditional perforations.

Another key innovation lies in the predictable geometric design of the BHA, which ensures controlled trajectory during drilling and enables precise re-entry into previously drilled channels — a feature rarely achievable with other radial systems. This allows for targeted re-treatments, chemical stimulations, or downhole measurements without compromising the channel integrity or needing further intervention to re-orient.

Stages of Mechanical Radial Drilling

Radial drilling process is executed in three key stages, each designed to ensure precise orientation, clean entry into the formation, and controlled drilling of radial channels from the main borehole. The process is highly repeatable and allows for accurate reservoir access with minimal deviation.

Running of the Anchor Module. The first stage involves the deployment and orientation of the anchor module (a sort of retrievable packer set below the zone of interest) at the target depth within the wellbore. Using gamma-ray logging, collar locator tools (caliper logging), and inclinometer data, the exact location of the desired reservoir interval is identified. The anchor module is securely installed in the borehole, ensuring stability and proper alignment for the upcoming radial drilling operation.

Milling Window (in Cased Wells). In the second stage, the milling BHA is run into the well and connects to the anchor module. This assembly mechanically mills a precise window or slot in the casing, creating an access point into the reservoir rock. Once the casing window is successfully cut, the milling BHA is retrieved to the surface, completing the access preparation phase.

Radial Channel Drilling. The final stage is the directional drilling of the radial channel. The drilling tool is lowered into the wellbore and engages the anchor module, guided by the integrated whipstock and fixed BHA geometry. Drilling proceeds along with the planned radial trajectory, creating a 2-3/4-inch diameter channel extending 45 to 80 feet into the reservoir. Upon completion of the radial bore, the drilling assembly is retrieved from the well.

Survey/Acid treatment/Screen Installation. In the same way how to perform drilling operation each drilled channel is accessible to perform any additional services like survey of trajectory, acid (any chemical) treatment or screens.

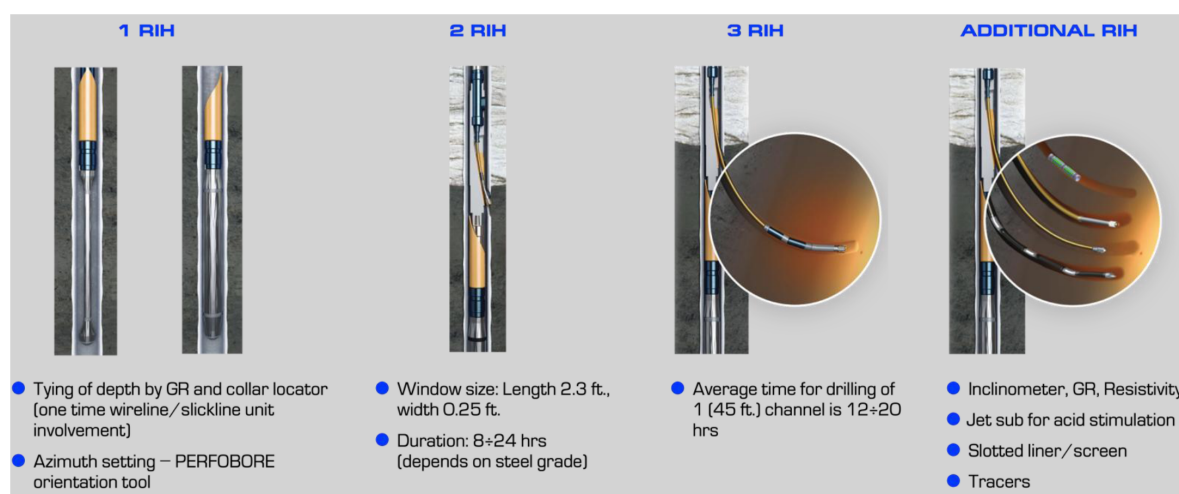


Figure 6—Sequence of operations while performing MRD

MRD technology, using conventional bits and small-sized mud motors, is adaptable to a wide range of wells — vertical, deviated, or horizontal — and is designed for efficiency, repeatability, and minimal disruption to existing infrastructure. Its success is not only rooted in technical innovation but also validated by its growing adoption across international oilfields, where it has been consistently linked to slower production decline rates and enhanced reservoir recovery when compared to traditional stimulation techniques.

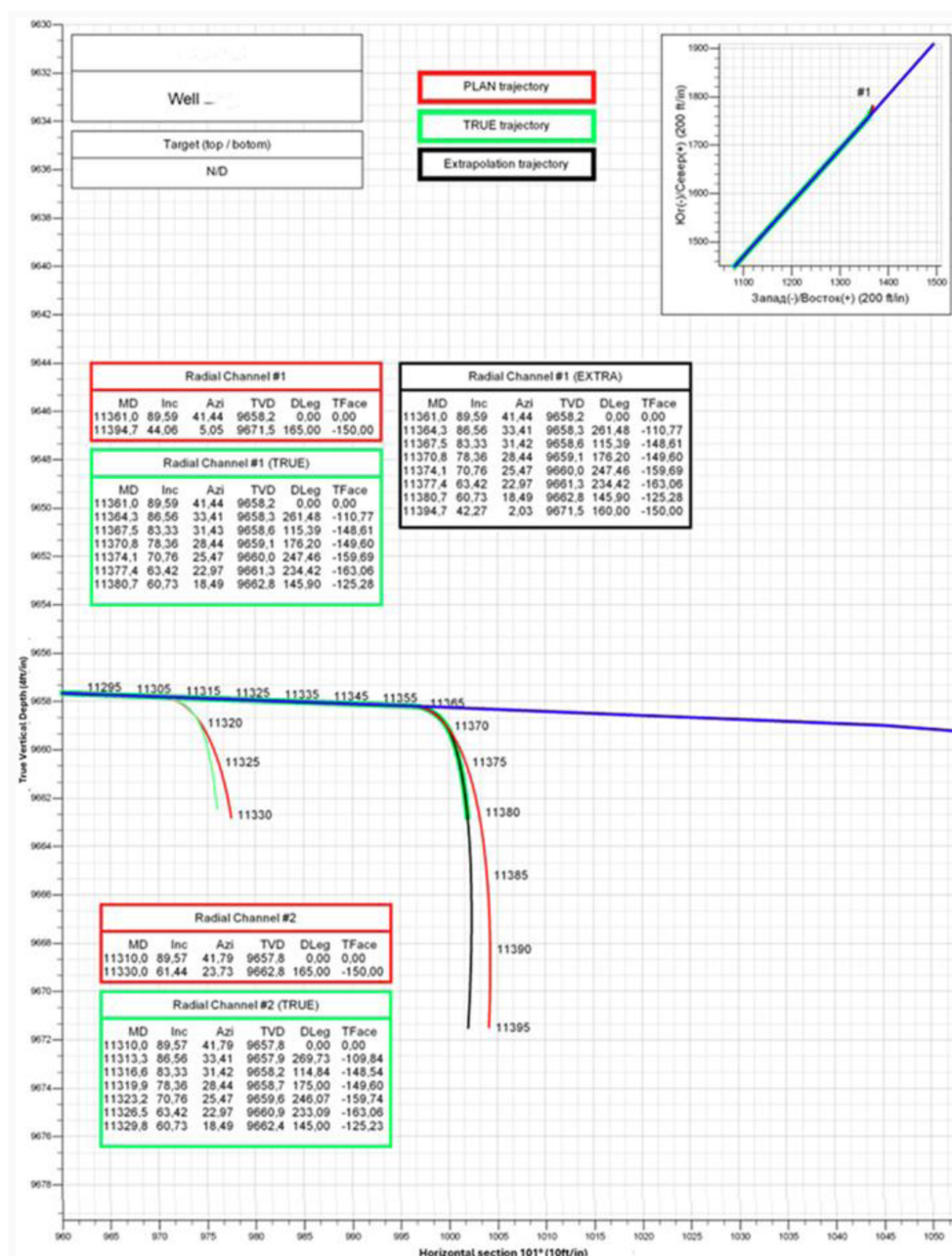
Drilling Assembly and Execution of Channel #1

Channel #1 was drilled with a 1-15/16" PDM and 4-bladed 8mm PDC bit, targeting depths of 11,361 ft to 11,394.7 ft. Pre-run checks confirmed equipment integrity. The assembly was space-out connected to the anchor, with connection verified through tensioning and rotational unloading, successfully releasing the inner string.

After drilling was completed, the well channel was flushed and the tool repositioned, with disconnection from the anchor achieved by applying 15 klbs over string weight and seven string turns. During shallow hole testing, thick mud with metallic debris was found in the BHA, but after cleaning and retesting, the assembly successfully ran in hole, the drilled path was recorded, and the assembly was retrieved.

Drilling Summary for Channel #2

The drilling of Channel #2 was performed achieving stable initial pressures and flow rates. Operations continued smoothly until 5.4 m depth, when a sudden pressure loss occurred and circulation could not be restored. Inspection revealed filter sub clogging and a broken nonmagnetic pipe, leaving a 4.1 m fish in the channel. The team opted not to re-drill Channel #2, instead proceeding with acidization and retrieval of Anchor 1. The total circulation time was 9.5 hours.



Acid Jetting Execution

Acid jetting operations were conducted on Channels #1 and #2 using 15% hydrochloric acid (HCl) delivered through jet subs equipped with four 8 mm bore inserts. For Channel #1, 15 barrels of acid were pumped after successful entry into the channel, resulting in a post-treatment injectivity of 2.7 BPM at 2,100 psi. The post-job inspection confirmed that all equipment was flushed, free of damage and corrosion, and remained in serviceable condition.

Channel #2 was treated with 30 barrels of acid. Similar injectivity results were observed. However, post-acidization, the well exhibited instability, including a kick at 9.0 ppg mud weight. Stabilization was achieved by displacing with 9.3 ppg mud. Subsequent deterioration in hole conditions, likely due to chemical interaction, raised the risk of stuck pipe and well control issues, including gas and oil at surface.



Figure 8—Jet Nozzle tool for acidizing

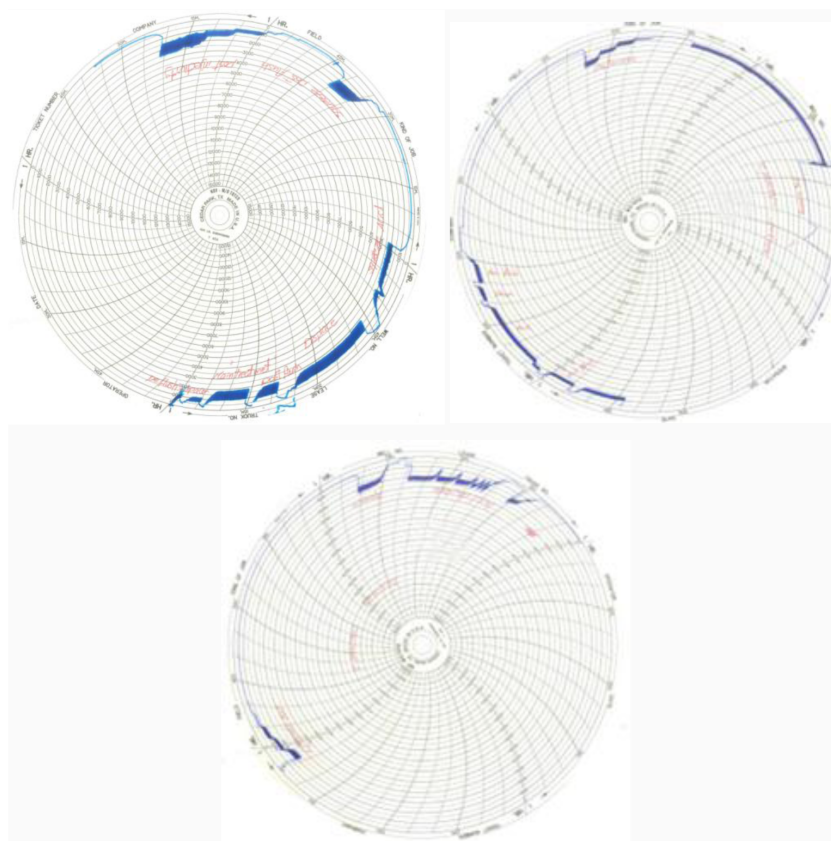


Figure 9—Acid pumping diagram for Channel-1 & 2 with Pre & Post Injection Profile

Channel #3 Drilling Attempt and Anchor Retrieval

Channel #3 was planned to target the interval from 11,116 ft to 11,136 ft. All system components passed pre-run inspections, and shallow hole pressure testing confirmed. During run-in-hole, the BHA failed to reach the anchor depth, stopping approximately 7 ft above the target despite unloading and circulation attempts. Upon pulling out, it was discovered that the gripper pins had sheared, and housing marks were present, indicating mechanical interference.

Subsequent attempts to flush and re-engage the anchor were met with similar obstruction, suggesting the anchor may have shifted upward. A dedicated anchor retrieval BHA was deployed, and the anchor was eventually released. However, significant overpulls (up to 45 klbs) and hang-ups were encountered during retrieval, and the string became stuck at 10,964 ft.

Attempts to free the string using annular pumping (160 gpm at 800 psi) and mechanical manipulation were unsuccessful. Acid was eventually pumped around the anchor and BHA to eliminate the seizure. This intervention proved effective, allowing partial recovery of the string. However, gas was observed at surface, prompting a decision to halt further acid pumping for safety reasons. Final retrieval of the anchor module was completed with intermittent hang-ups up to 34 klbs. Post-operation inspection revealed that the anchor was not fully deactivated, and the sludge collector was completely blocked.

Operational Summary of well ABC Radial Drilling and Acid Jetting

The Well ABC presented a technically demanding open-hole horizontal environment, characterized by high temperatures and depth that challenged conventional survey tools. Stimulation introduced a high-temperature survey tool compatible with the drilling bottom hole assembly (BHA), along with an upgraded swivel—introduced based on prior experience in the XYZ field—to manage the required string rotation under load needed to manage the drag in the horizontal wellbore and weight manipulation at the bottom hole.

The use of a Kelly rig imposed significant limitations on channel length and string manipulation during run-in-hole (RIH) operations. The rig's minimum rotary speed of 20 RPM exceeded the optimal range recommended by Stimulation (5–10 RPM), complicating control over drilling parameters. Early trips were delayed, and a conventional survey tool failed due to premature activation and thermal collapse. Orientation was later captured using the new survey tool, with a backup deployed within 24 hours.

Surveying Channel #1 was successful, though magnetic interference and short channel length limited azimuth reliability. Inclinator data from two stations was sufficient to verify trajectory. The wellbore initially allowed smooth tripping, but conditions deteriorated toward the end of operations, likely due to acid interaction—complicating BHA handling and increasing stuck-pipe risk.

Acid jetting was performed using 15% HCl. Channel #1 received 15 bbl, achieving 2.7 BPM at 2,100 psi. Channel #2 was treated with 30 bbl, yielding similar injectivity. However, post-acidization, the well exhibited instability, including gas and oil at surface and fluid losses at 9.0 ppg mud weight. Stabilization required displacement with 9.3 ppg mud and conditioning run. Acid was also used to free a stuck anchor.

Due to deteriorating hole conditions and increasing operational risk, the decision was made to complete the well after Channels #1 and #2, foregoing Channels #3 and #4. The Channel #3 attempt was aborted due to mechanical interference and anchor retrieval complications, which required acid intervention to resolve the issue. The anchor was eventually recovered, though the sludge collector was found fully blocked and the anchor not fully deactivated.

This operation demonstrated the limitations of Kelly rigs for Stimulation applications, particularly in managing channel length and downhole equipment manipulation. Future deployments should consider alternative rig configurations to optimize performance and reduce operational risk.

Novel Concepts

The novel concepts and advanced techniques in drilling 2-3/4" channels have significantly improved the efficiency and precision of mechanical radial drilling operations. By integrating high-precision equipment and directional drilling techniques, operators can create stable channels that meet the required specifications. Additionally, the use of drilling muds and reinforcement techniques ensures that the channels remain stable during reentry activities. These innovations not only enhance the safety and reliability of drilling operations but also pave the way for more complex and ambitious mechanical radial drilling projects in the future.

Results

The recent implementation of stimulation technology in the XYZ field has yielded significant results. This marked the first successful application of such technology in the area. Two radial channels were drilled with precise control over their trajectory, ensuring a successful connection to the main wellbore. The operation achieved an average rate of penetration (ROP) of 3 feet per hour, surpassing the previous benchmark of 2 feet per hour set during the previous operations.

Both channels were subsequently re-entered and treated with acid using a specialized jetting assembly. The acid treatments proved effective, enhancing both injectivity and stimulation within the well.

Notably, there were no Health, Safety, and Environmental (HSE) incidents recorded or attributed to the stimulation operations. This speaks to the thorough planning and execution of the project. The stimulation system was successfully operated in horizontal open-hole conditions using a Kelly rig. Despite the rig's limitations, channel lengths were maximized through precise space-out and anchor placement.

A modified lock nut was deployed for the first time, which reduced the need for anchor flushing. Additionally, a high-temperature survey tool, capable of operating reliably at temperatures exceeding 130°C, provides essential trajectory data. This tool was successfully run inside the drilling Bottom Hole Assembly (BHA) for a washdown survey, marking its first-time implementation.

Upgraded swivels performed effectively under conditions of high drag and restricted rig parameters. Two open-hole anchors were set and retrieved without any incidents. Lastly, a stuck-pipe event was managed successfully through acid intervention, preventing any significant delays or complications.

These results highlight the advancement and effectiveness of stimulation technology, demonstrating its potential for future applications in similar fields.

Conclusion

The post-stimulation intervention at well ABC demonstrated a significant enhancement in well productivity, as validated by two consecutive MPFM tests conducted on 25th and 26th July 2025. Compared to the initial oil production rate of X BOPD, the well achieved Y BOPD on 25 July and Z BOPD on 26 July, reflecting a 35.11% and 98.50% increase respectively.

This uplift was accompanied by a substantial reduction in water production and a more stabilized gas-oil ratio, indicating improved reservoir connectivity and effective cleanup of the near-wellbore region. The consistent separator conditions and absence of diagnostic alarms further affirm the reliability of the test data.

These results underscore the operational viability and production impact of radial drilling in mature carbonate reservoirs. Success at well ABC supports broader deployment of radial drilling and targeted stimulation strategies across similar assets within the Operator Company portfolio.

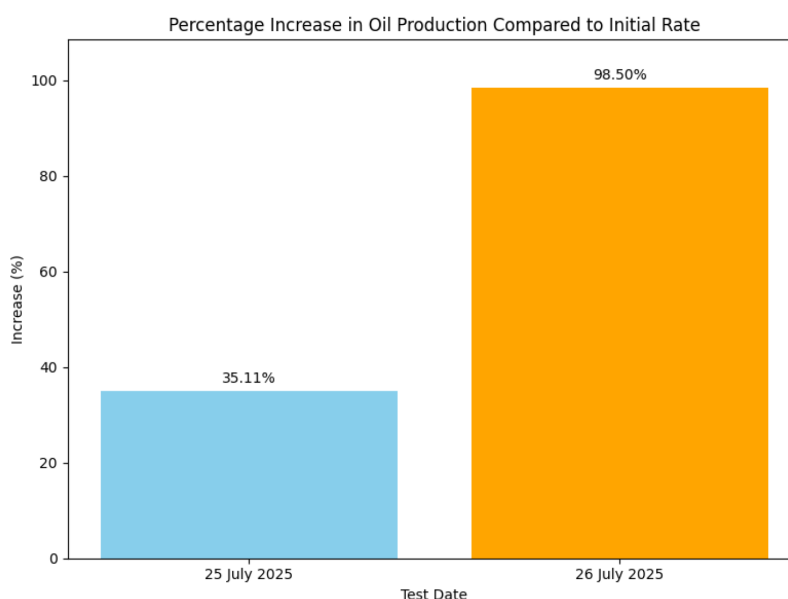


Figure 10—Oil Production Increase (2X compared to initial well production)

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